PERMANENCE OF SMOOTH/\'ETALE ALONG A SURJECTIVE FLAT LFP F

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites three Morphisms of Spaces lemmas and one Descent lemma, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- \'Etale locality of smooth/\'etale morphisms. Stacks Project, Morphisms of Spaces, Lemmas spaces-morphisms-lemma-smooth-local and spaces-morphisms-lemma-etale-local. Justifies the reduction to affine B and Y.
- Openness of fppf morphisms on points. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-fppf-open. Lets the proof pick U with $U \to Y$ surjective.
- Scheme-side permanence. Stacks Project, Descent, Lemma descent-lemma-smooth-permanence. Concludes for the scheme diagram.

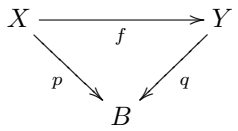
Source reference. Original reference 03ZF: sources/stacks-chow/tags.txt, source lines 5218–5218. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03XT: sources/stacks-chow/tags.txt, source lines 5160–5160. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 042S: sources/stacks-chow/tags.txt, source lines 5339–5339. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 05B5: sources/stacks-chow/tags.txt, source lines 6873–6873. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let*



be a commutative diagram of morphisms of algebraic spaces over S. Assume that

(1) *f is surjective, flat, and locally of finite presentation,*
(2) *p is smooth (resp. étale).*

Then q is smooth (resp. étale).

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Proof. We deduce this from the analogue for schemes. Namely, the problem is étale local on B and Y (Morphisms of Spaces, Lemmas 03ZF and 03XT). Hence we may assume that B and Y are affine schemes. Since $|X| \to |Y|$ is open (Morphisms of Spaces, Lemma 042S), we can choose an affine scheme U and an étale morphism $U \to X$ such that the composition $U \to Y$ is surjective. In this case the result follows from Descent, Lemma 05B5. □